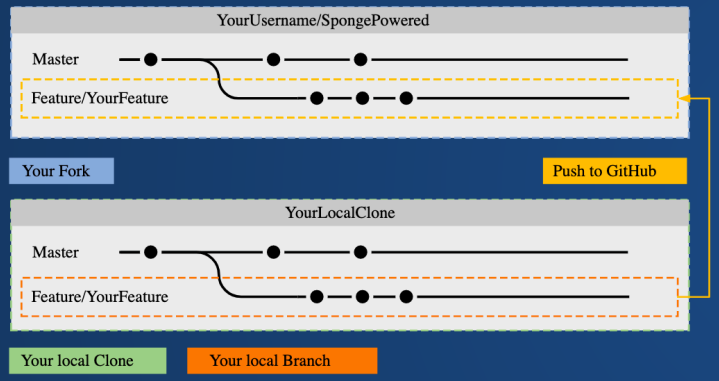
Software Quality and Project Development

Università degli Studi di Catania
Dipartimento di Matematica e Informatica



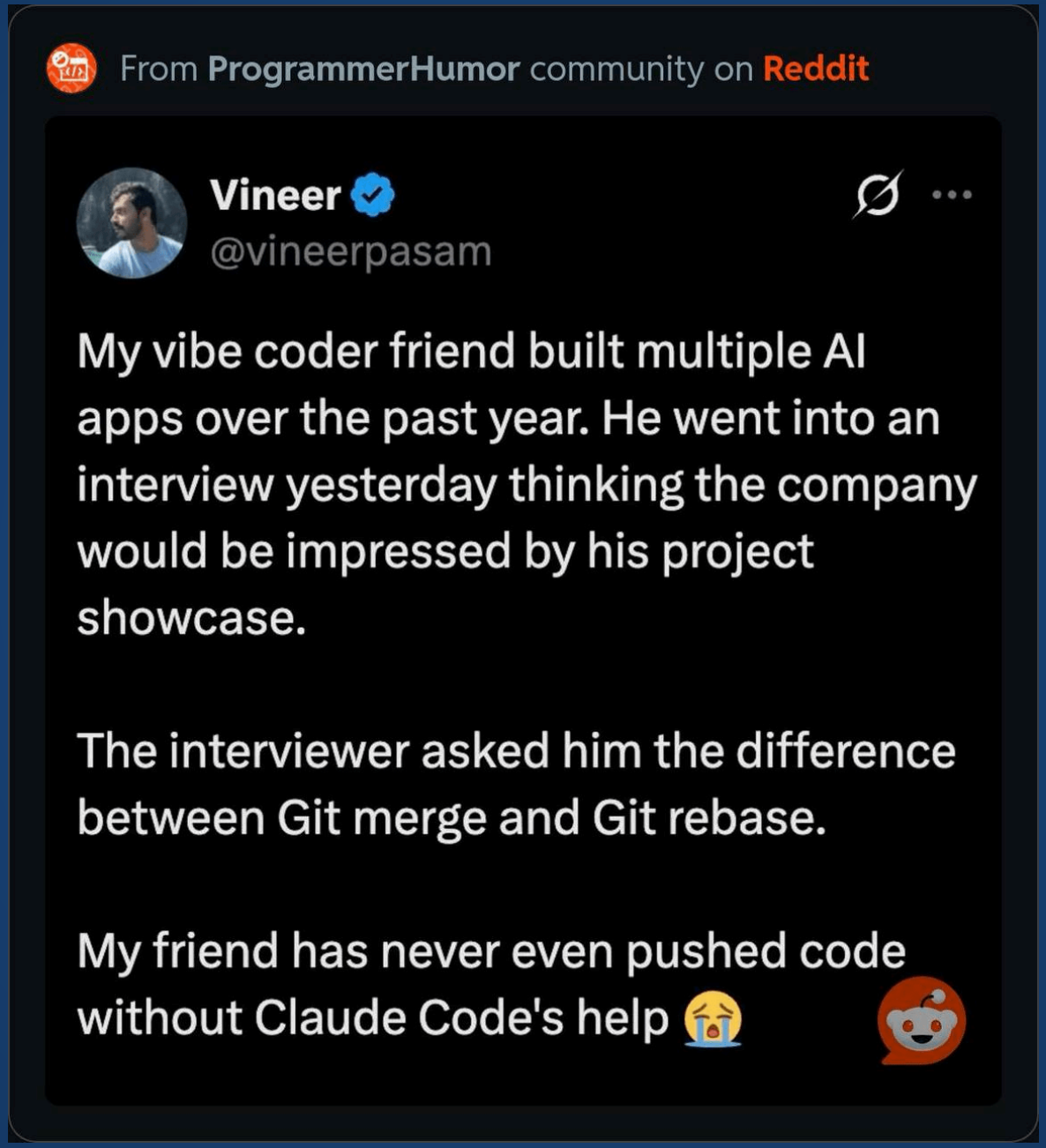


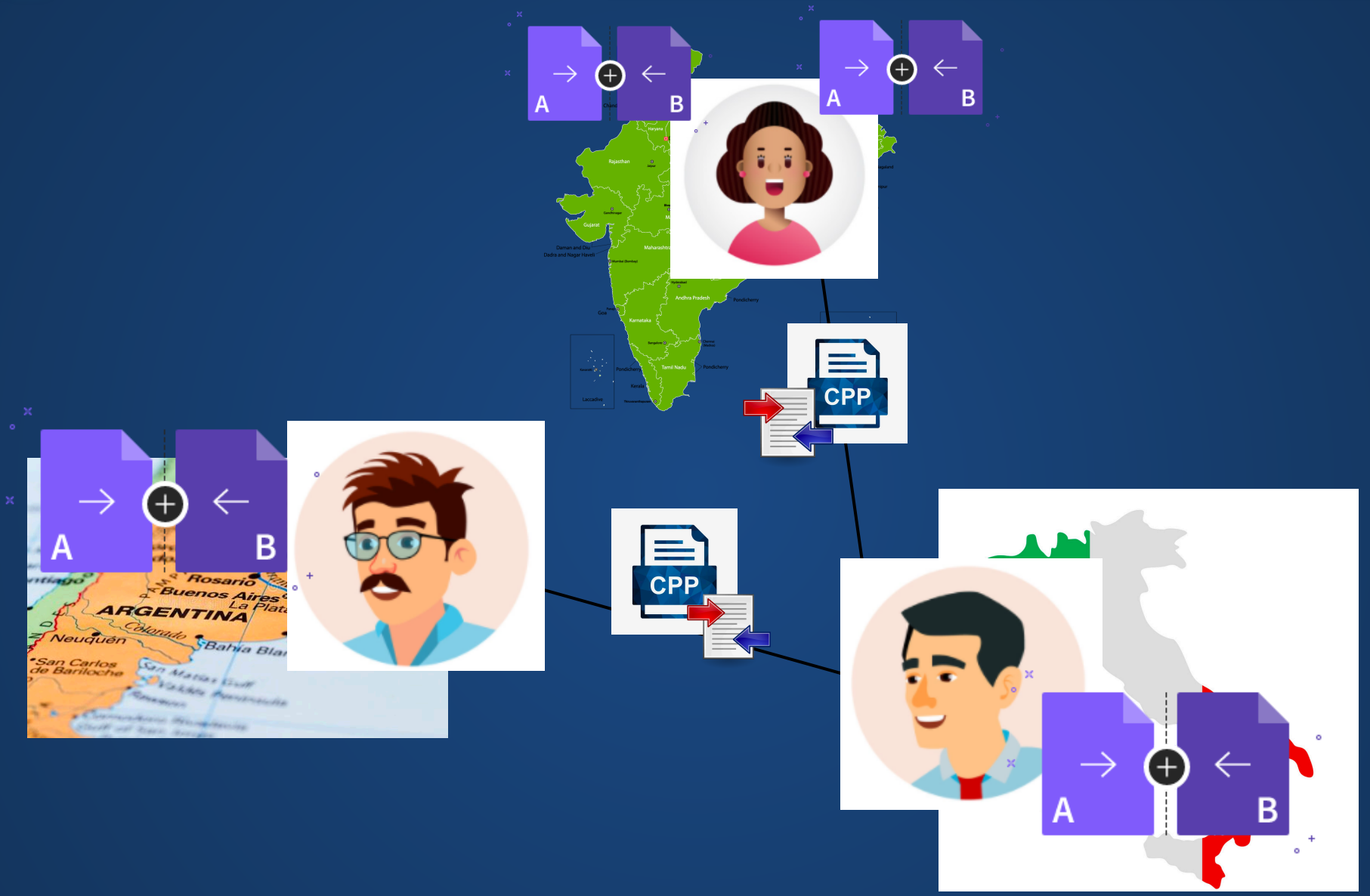
Git

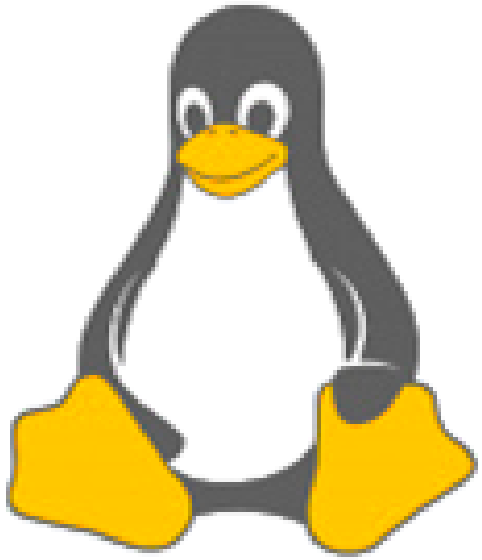
A way to manage your project



2005


From ProgrammerHumor community on Reddit
Vineer
@vineerpasam
My vibe coder friend built multiple AI apps over the past year. He went into an interview yesterday thinking the company would be impressed by his project showcase.
The interviewer asked him the difference between Git merge and Git rebase.
My friend has never even pushed code without Claude Code's help


A
B
A
B
CPP
A
B
CPP
ARGENTINA
Rosario
Buenos Aires
Neuquén
A
B



Linux™

git

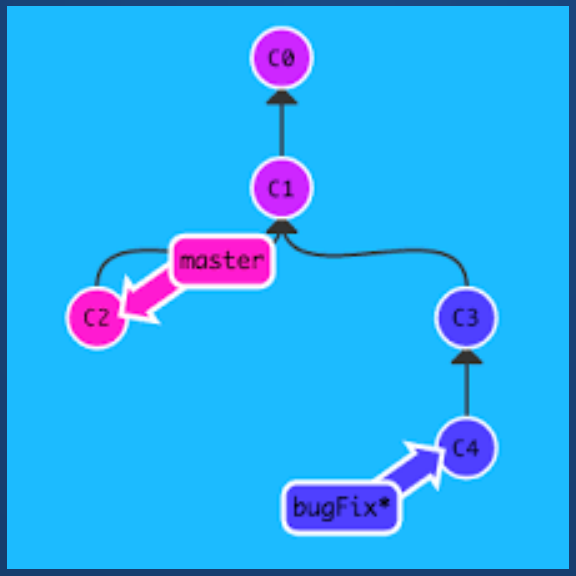
https://ohmygit.org/



https://github.com/firstcontributions/first-contributions



Learn to make your first open source contribution on GitHub in 5 minutes



https://learngitbranching.js.org/

```
$ git config

$ git add
$ git rm
$ git mv
$ git commit -m 'desc commit'

$ git checkout -b branch_name # craete a new branch
$ git checkout branch_name
$ git merge REMOTE BRANCH # ex. git merge origin master

$ git reset
$ git revert

$ git status
$ git log
$ git diff

$ git init
$ git clone
$ git remote → git remote add, git remote -v, git remote rm
$ git fetch

$ git pull
$ git push
```

How to install GIT



Linux (or WSL), via package manager:

```
$ apt install git
```



Mac via Homebrew or MacPort:

```
$ brew install git
$ port install git
```



Windows via winget

```
$ winget install --id Git.Git -e --source winget
```

or https://gitforwindows.org/

```
1
2  $ git config --global user.name "Stefano Borzì"
3  $ git config --global user.email "stefano@example.com"
4  $ git config --global core.editor nano
5
6
7  $ git config --list
8  user.name=Stefano Borzì
9  user.email=stefano@example.com
10 core.editor=nano
11 ...
12
13
14 $ git config user.name
15 Stefano Borzì
16
```

```
$ git init

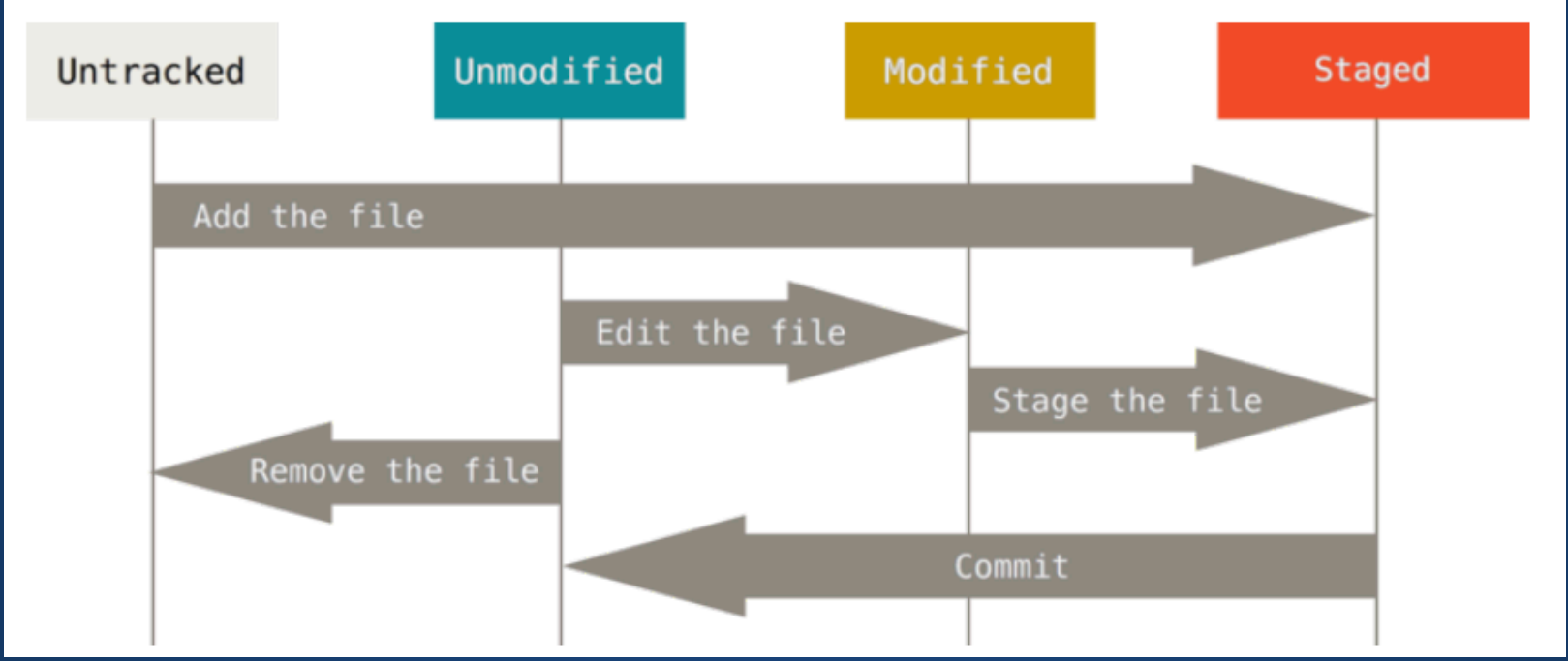
$ git add
$ git rm
$ git mv

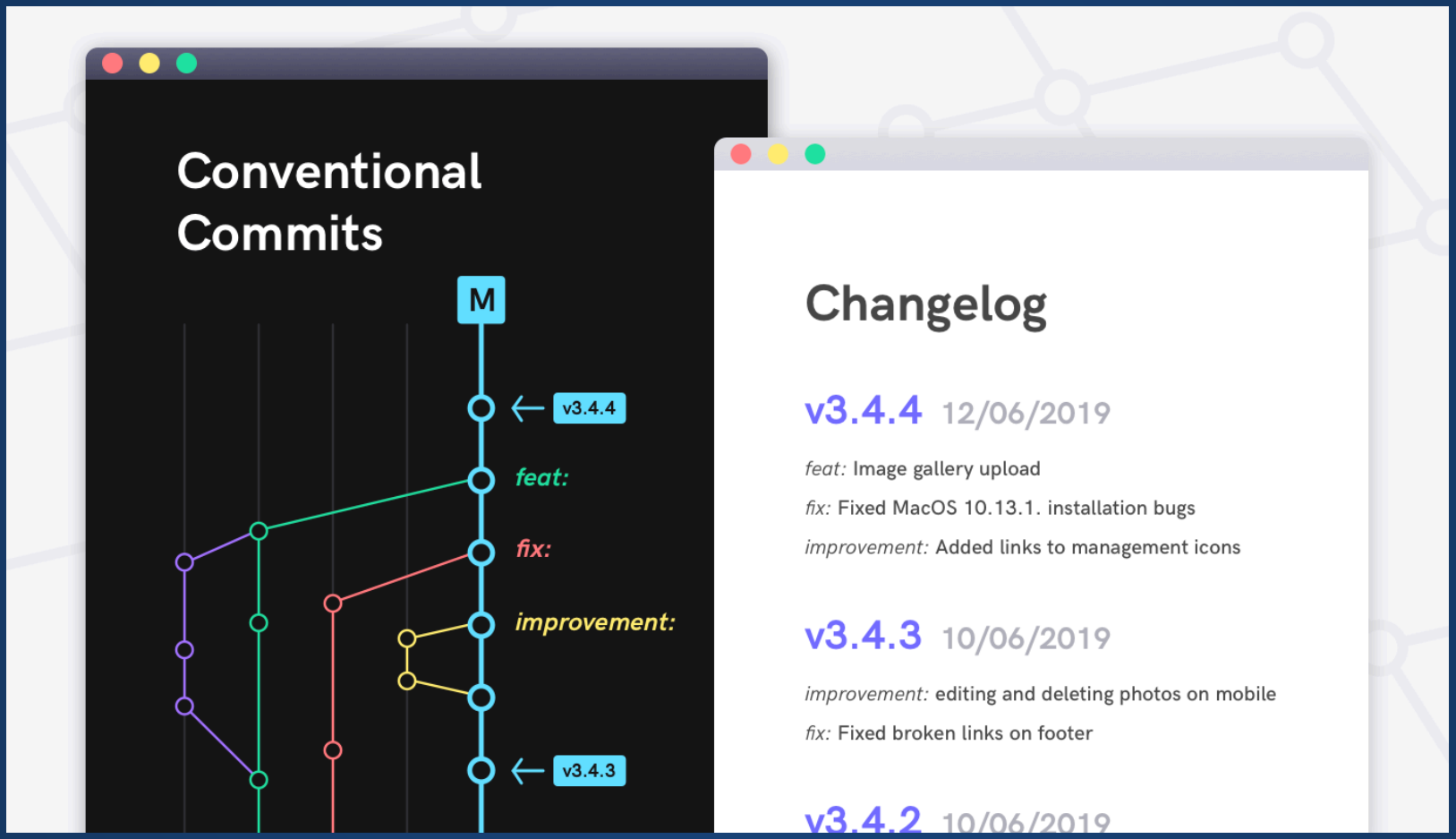
$ git status
$ git diff

$ git commit -m 'desc commit'

$ git log
```

git status

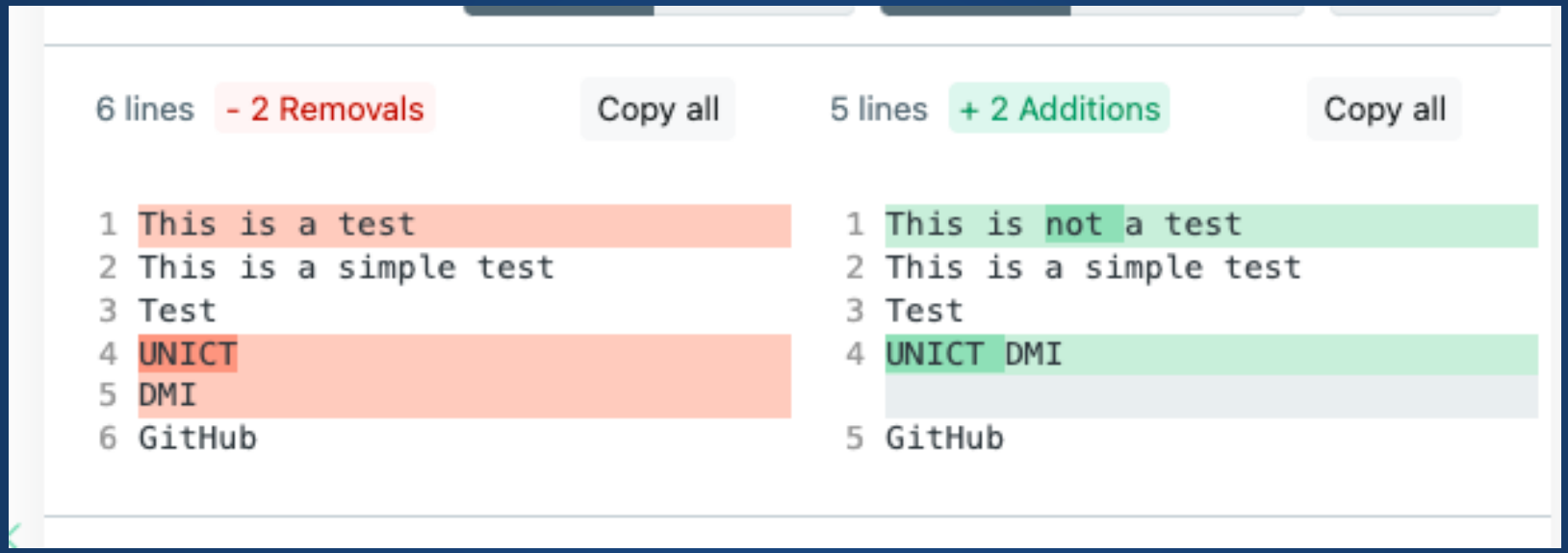

Untracked
Unmodified
Modified
Staged
Add the file
Edit the file
Stage the file
Remove the file
Commit


Conventional Commits
M
v3.4.4
feat:
fix:
improvement:
v3.4.3
Changelog
v3.4.4 12/06/2019
feat: Image gallery upload
fix: Fixed MacOS 10.13.1. installation bugs
improvement: Added links to management icons
v3.4.3 10/06/2019
improvement: editing and deleting photos on mobile
fix: Fixed broken links on footer
v3.4.2 10/06/2019

Conventional Commits

```
1 fix: for a fix                        ex. fix(main): windows build
2 feat: implementing a new feature      ex. feat(home): add footer
3 docs: for documentation.              ex. doc(contribution): add contribution guidelines
4 refactor: for refactoring purposes    ex. refactor(tests): replace all "pippo" variables
5 test: adding unit-tests, e2e-etest    ex. test(lessons): add unit-tests for lesson
6 chore: minor improvements             ex. chore(merge): solve conflict
```

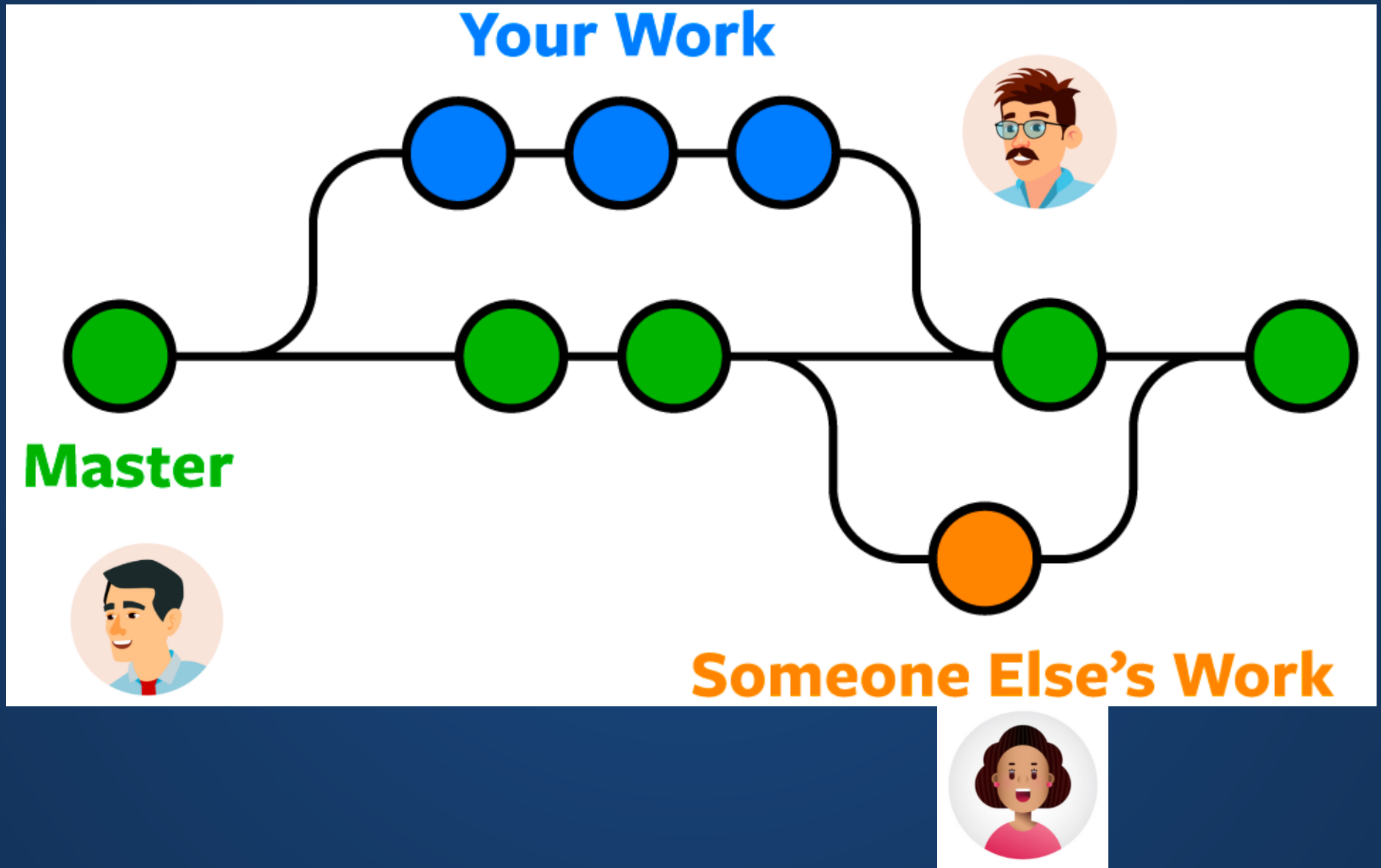
git diff


6 lines
- 2 Removals
Copy all
1 This is a test
2 This is a simple test
3 Test
4 UNICT
5 DMI
6 GitHub
5 lines
+ 2 Additions
Copy all
1 This is not a test
2 This is a simple test
3 Test
4 UNICT DMI
5 GitHub

checkout, revert, reset

```
1 $ git checkout [COMMIT]
2
3 $ git revert [COMMIT]
4
5 $ git reset
```

Branches and Merges



Branches and Merges

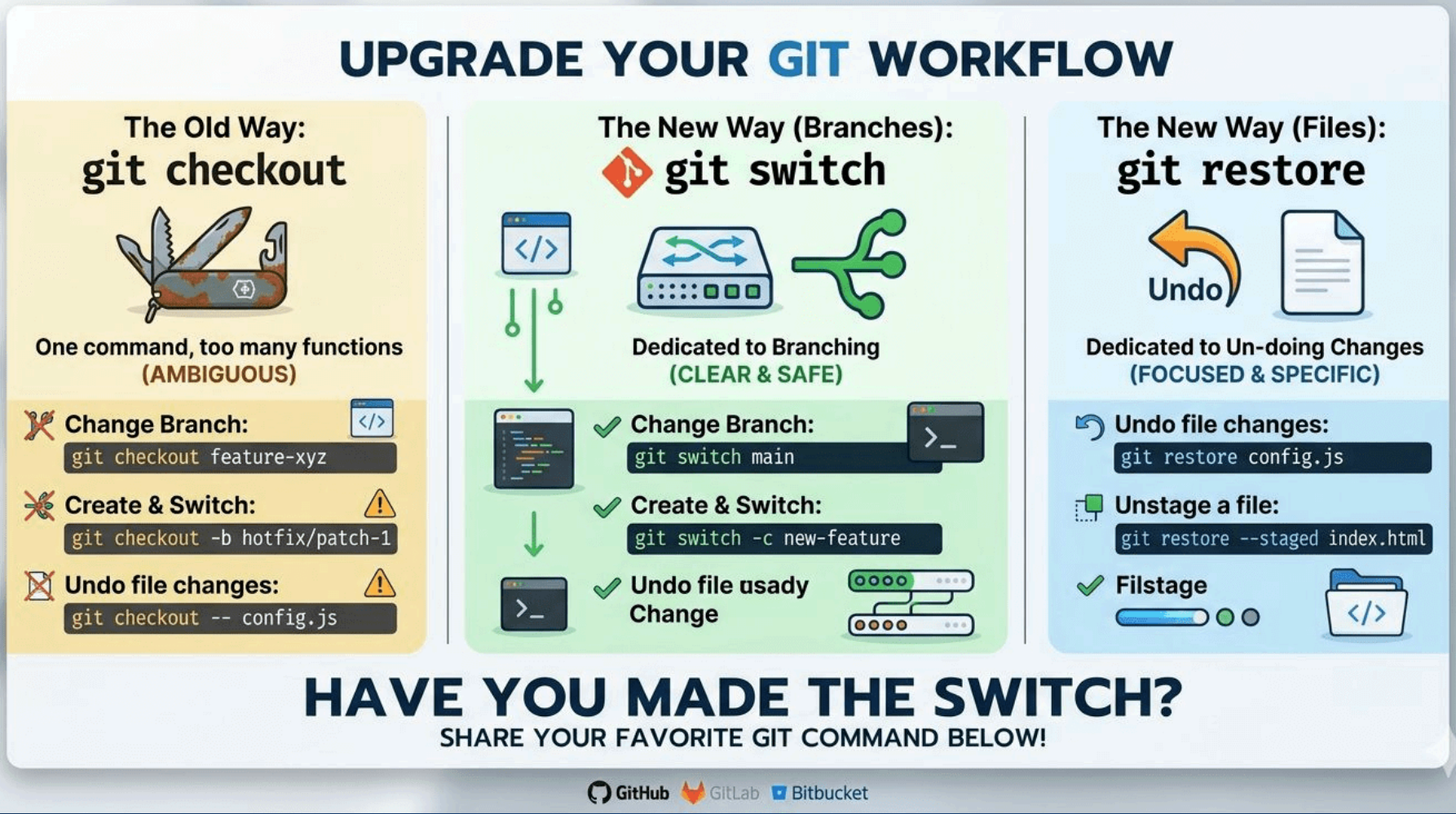
```
1  # craete a new branch and checkout
2  $ git switch -c branch_name
3  $ git checkout -b branch_name
4
5
6  $ git switch branch_name
7  $ git checkout branch_name
8
9  $ git merge [REMOTE] BRANCH
10 # ex. git merge origin master
```

Restore files

```
$ git restore fileName.cpp

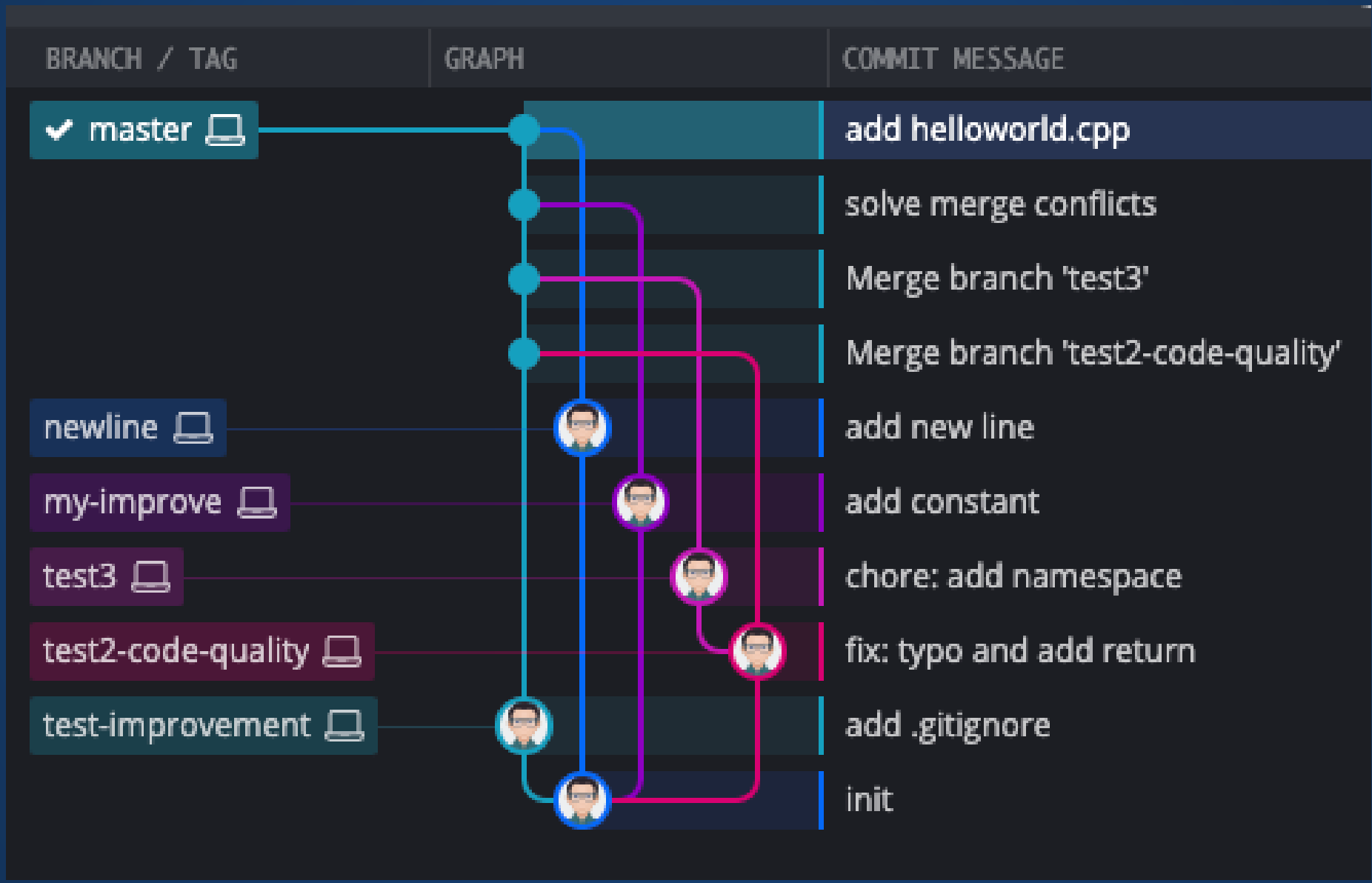
$ git checkout fileName.cpp
```

Branches and Merges

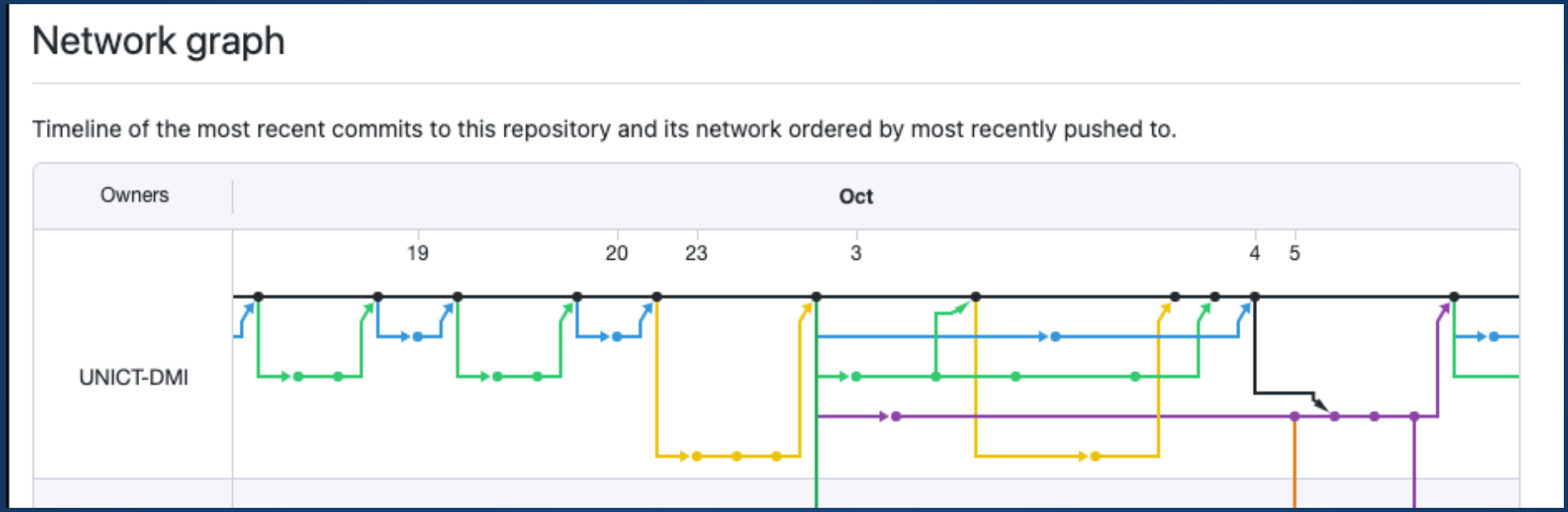


https://www.linkedin.com/posts/roberto-heckers-2313453b_stop-using-git-checkout-for-everything-it-activity-7440797820653211648-SNHV/

Branches and Merges



DMI Bot - network graph



https://github.com/UNICT-DMI/Telegram-DMI-Bot/network

.gitignore

https://git-scm.com/docs/gitignore

```
# See http://help.github.com/ignore-files/ for more about ignoring files.

# compiled output
/dist
/tmp
*.js.map

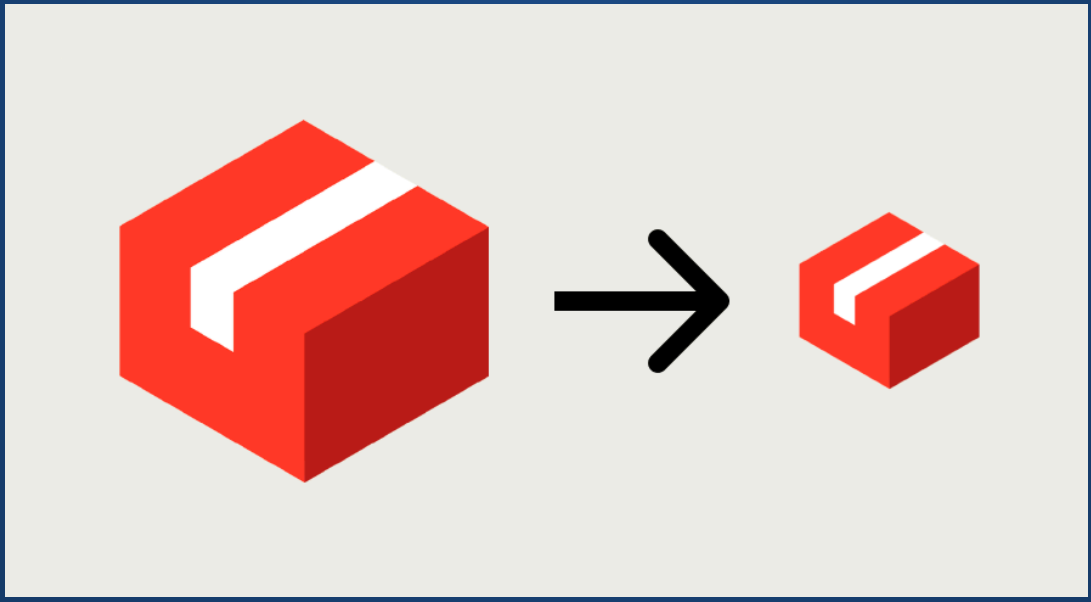
# dependencies
/node_modules

# IDEs and editors
/.idea
.project
.classpath
.c9/
*.launch
.settings/
*.sublime-workspace
.vscode/*

# System Files
.DS_Store
Thumbs.db
```

GIT LFS





GIT LFS

https://speakerdeck.com/player/24c9d09fdccd43b8aa397b254502a810

Visual Studio Code plugins

Git related

- Git Lens
- Git History

Generic plugins

- sonarlint
- change-case



Exercise

$ wget https://bit.ly/3ucdNHK

```
/*
Make a BRANCH per each task.

- fix warning and errors
- fix the code
- make N dynamic, let the user choose N
- remove unused
- make some improvements on your own
*/
```

all the commands shown so far

```
$ git config

$ git init

$ git add
$ git rm
$ git mv
$ git commit -m 'desc commit'

$ git checkout -b branch_name # craete a new branch
$ git switch -c branc_name

$ git checkout branch_name
$ git switch branch_name

$ git merge BRANCH # ex. git merge feature-1

$ git restore
$ git reset
$ git revert

$ git status
$ git log
$ git diff
```